

COMP1531

Software Engineering Fundamentals

Week 7

Design - Conceptual Modelling

In this Lecture:

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- Describe **conceptual modelling** and differentiate it from physical models
- Identify common **types of conceptual models** used in software design
- Explain why models are essential for **communication** between stakeholders
- Recognise **state diagrams** and describe how they represent system behaviour



What is a Model?

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What is a Model?

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What is a Model?

Figure 1: Mathematical Model

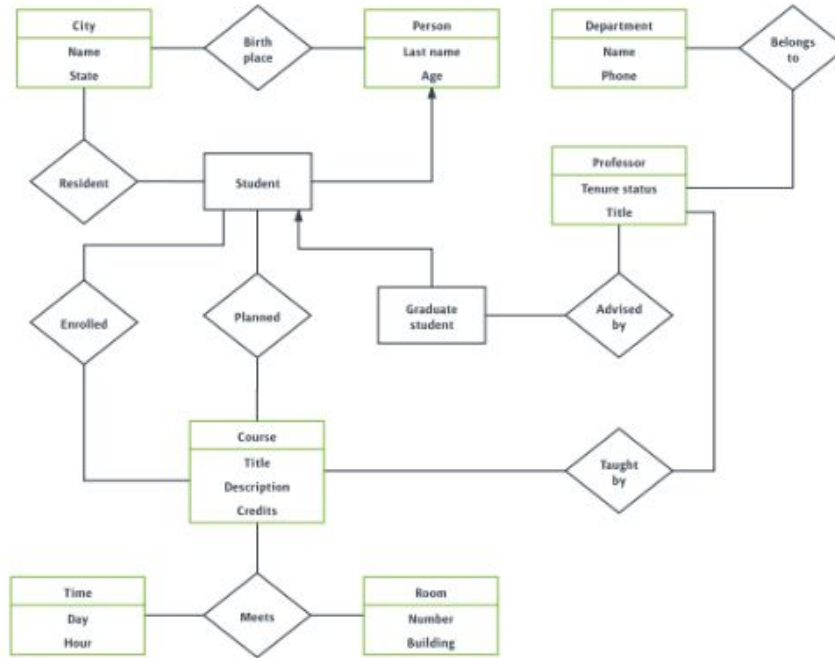
$$\begin{aligned}
 \frac{dC}{dt} &= \frac{s_c C}{1 + (R_b + N_b + T_b + L_b) / b_{cm}} + \frac{k_{cp}}{v_m} (P - C) - (k_{crc} + k_{cmc} + k_{ctc} + k_{clc}) C - \mu_c C \\
 \frac{dP}{dt} &= \frac{k_{cp}}{v_b} (C - P) - \mu_p P \\
 \frac{dR_c}{dt} &= \frac{s_{rc} R_c}{1 + R_b / b_{rm}} + k_{crc} C - k_{rcm} R_c - \mu_{rc} R_c & \frac{dR_m}{dt} &= k_{rcm} R_c - \frac{k_{rmb}}{v_m} R_m - \mu_{rm} R_m \\
 \frac{dN_c}{dt} &= \frac{s_{nc} N_c}{1 + N_b / b_{nm}} + k_{cmc} C - k_{ncm} N_c - \mu_{nc} N_c & \frac{dN_m}{dt} &= k_{ncm} R_c - \frac{k_{nmb}}{v_m} N_m - \mu_{nm} N_m \\
 \frac{dT_c}{dt} &= \frac{s_{tc} T_c}{1 + T_b / b_{tm}} + k_{ctc} C - k_{tcm} T_c - \mu_{tc} T_c & \frac{dT_m}{dt} &= k_{tcm} T_c - \frac{k_{tmb}}{v_m} T_m - \mu_{tm} T_m \\
 \frac{dL_c}{dt} &= \frac{s_{lc} L_c}{1 + L_b / b_{lm}} + k_{clc} C - k_{lcm} L_c - \mu_{lc} L_c & \frac{dL_m}{dt} &= k_{lcm} L_c - \frac{k_{lmb}}{v_m} L_m - \mu_{lm} L_m \\
 \frac{dR_b}{dt} &= \frac{k_{rmb}}{v_m} R_m - \mu_{rb} R_b & \frac{dT_b}{dt} &= \frac{k_{tmb}}{v_m} T_m - \mu_{tb} T_b \\
 \frac{dN_b}{dt} &= \frac{k_{nmb}}{v_m} N_m - \mu_{nb} N_b & \frac{dL_b}{dt} &= \frac{k_{lmb}}{v_m} L_m - \mu_{lb} L_b
 \end{aligned}$$

What is a Model?

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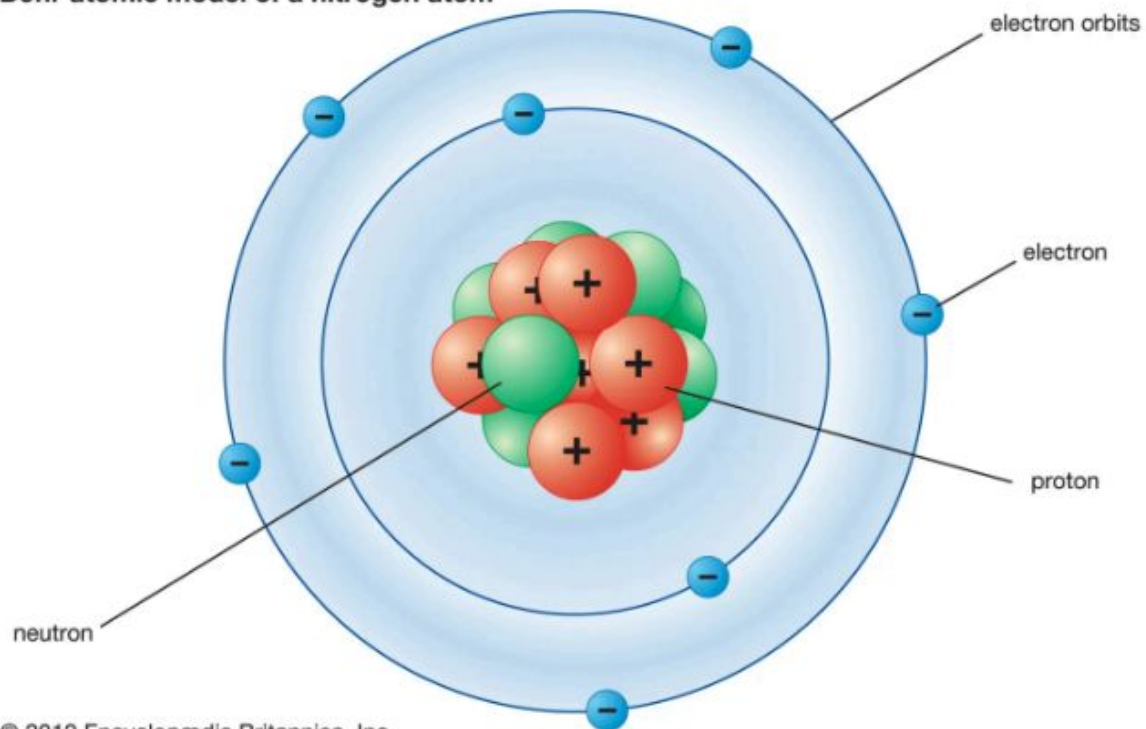


What is a Model?

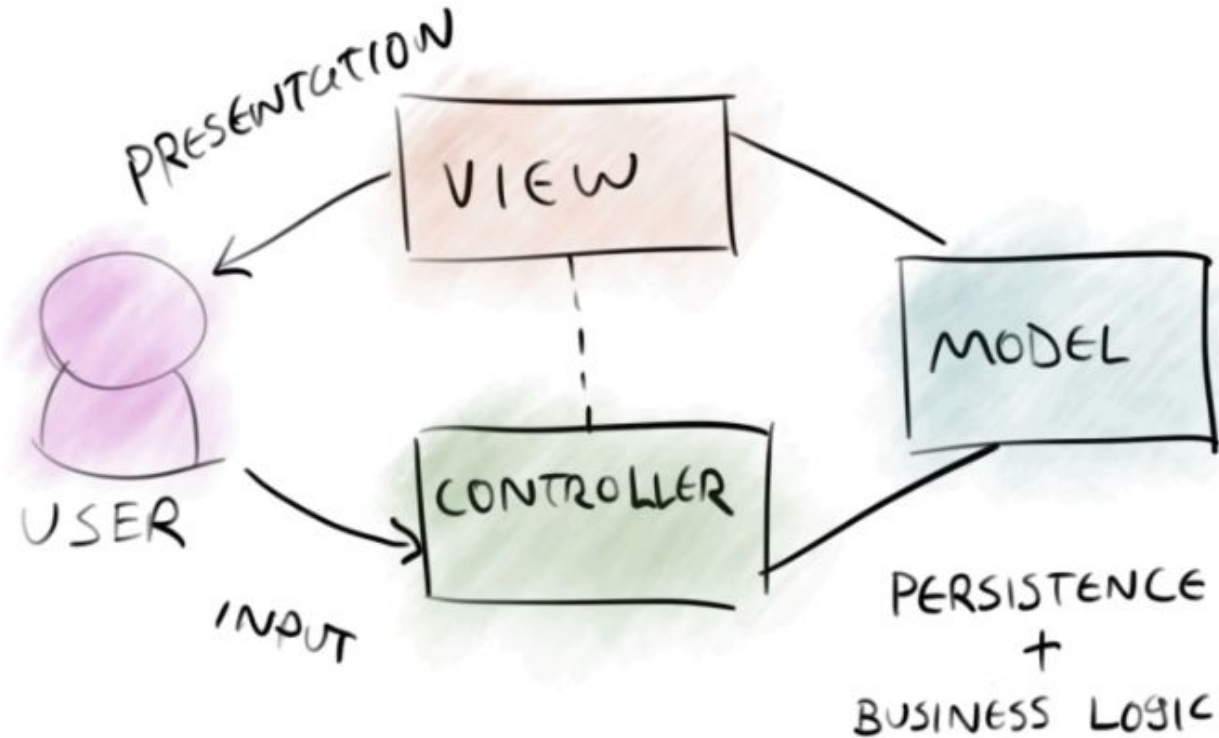


What is a Model?

Bohr atomic model of a nitrogen atom



What is a Model?





What is a Model?

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A model is a **simplified representation** of a more complex system that helps us understand, communicate, or reason about that system.

Key Points

- Reduces complexity → focuses on important features
- Helps predict, analyse, or document system behaviour
- Used across many domains - from maths to software to aerodynamics

Why models matter

- Enable communication between stakeholders
- Provide a reference for design and implementation
- Support reasoning about current, past, or future system states
- Models are cheaper to create than building real systems.



Conceptual Modelling

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A conceptual model is a type of model that captures a system in a conceptual way rather than a physical way.

They tend to be **diagrammatic** or **visual**.



Conceptual Modelling



Structural vs Behavioural Models

Structural Models

Focus on the **static structure** of the system:

- Objects, data, entities, and relationships
- What components exist

Examples

- UML Class Diagrams (Object Oriented Programming)
- ER Diagrams (Databases)

Use When

- Designing system architecture
- Defining data structures and relationships



Conceptual Modelling



Structural vs Behavioural Models

Behavioural Models

Focus on the **dynamic behaviour** of the system:

- How the system changes over time
- Events, actions, and transitions

Examples

- State Diagrams (State Machines)
- Use Case Diagrams (User flows)

Use When

- Describing workflows or user interaction
- Modelling logic and system responses



What Are Models Used For?

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Models help us:

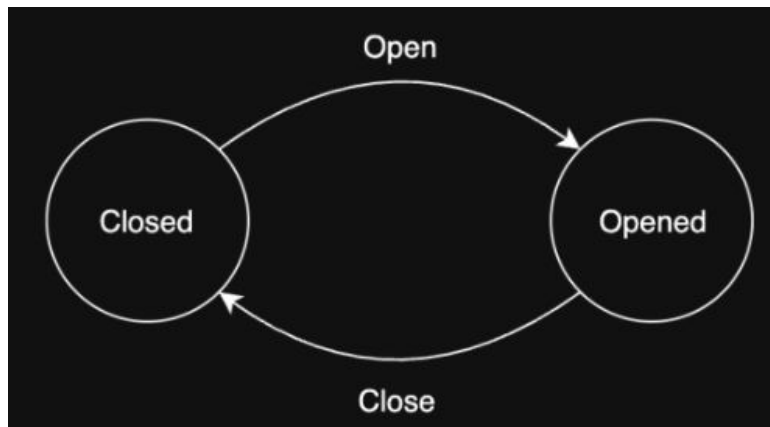
- **Understand the current system**
 - E.g., diagramming an existing login process to identify issues
- **Predict future behaviour**
 - E.g., modelling load on a server after user growth
- **Explore “what-if” scenarios**
 - E.g., how a UI behaves when network fails
- **Support communication**
 - E.g., explaining system design to teammates using UML
- **Document system knowledge**
 - E.g., storing state diagrams for future developers



State Diagrams

A **behavioural model** showing:

- Distinct **states** a system can be in
- **Transitions** between states triggered by events/actions
 - Simple example: a door
- Some variation in notation, though typically: states are circles, transitions are labelled arrows connecting them



States

- Closed
- Open

Transitions

- **open action** (Closed → Open)
- **close action** (Open → Closed)



State Diagrams

Suppose a door has the following states and transitions. Draw a **revised** state-transition diagram for the door scenario.

States

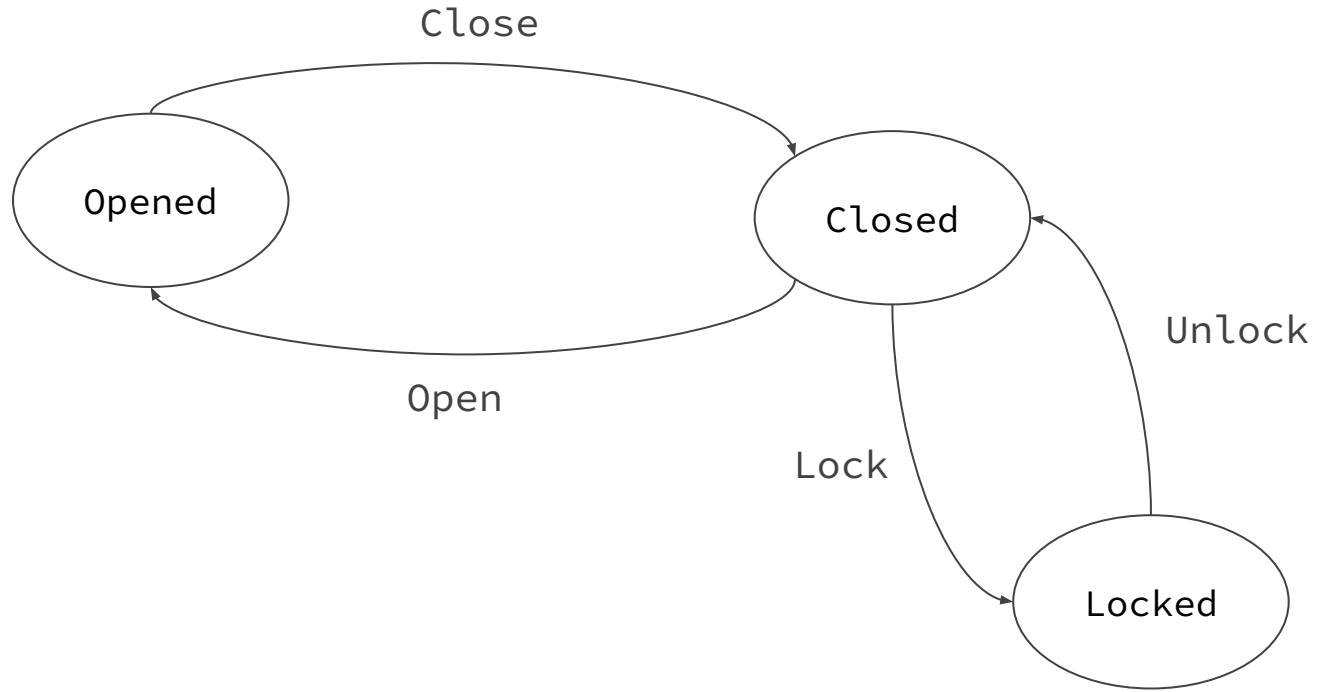
- Closed
- Open
- Locked

Transitions

- open action
- close action
- lock action
- unlock action



Answer:





State Machines

What is a State Machine?

- A computational model that consists of:
 - A **finite set of states**
 - **Transitions** triggered by events or actions
 - **Rules** defining how the system moves between states

Examples in Software:

- UI screen transitions
- Login/logout flows
- Network protocols (e.g., TCP states)
- Game character states (idle, running, jumping)

Parking Meter Example - State Machine

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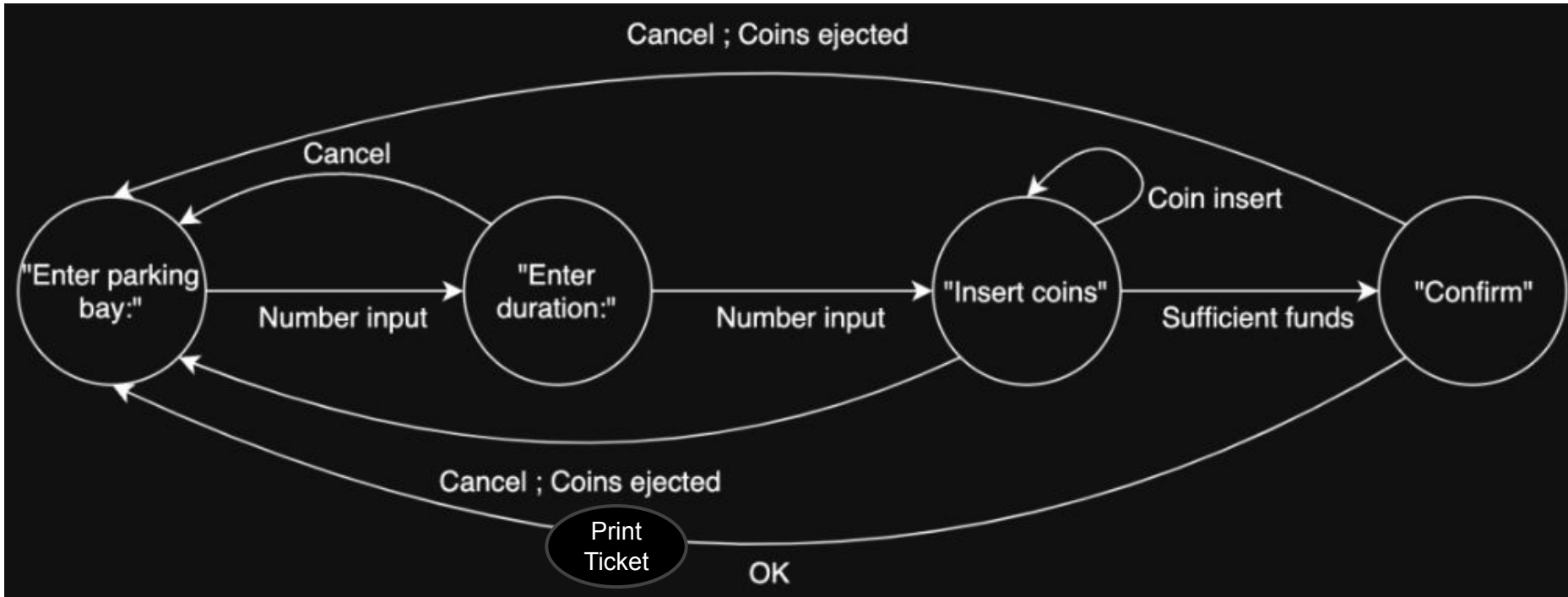


Parking Meter States



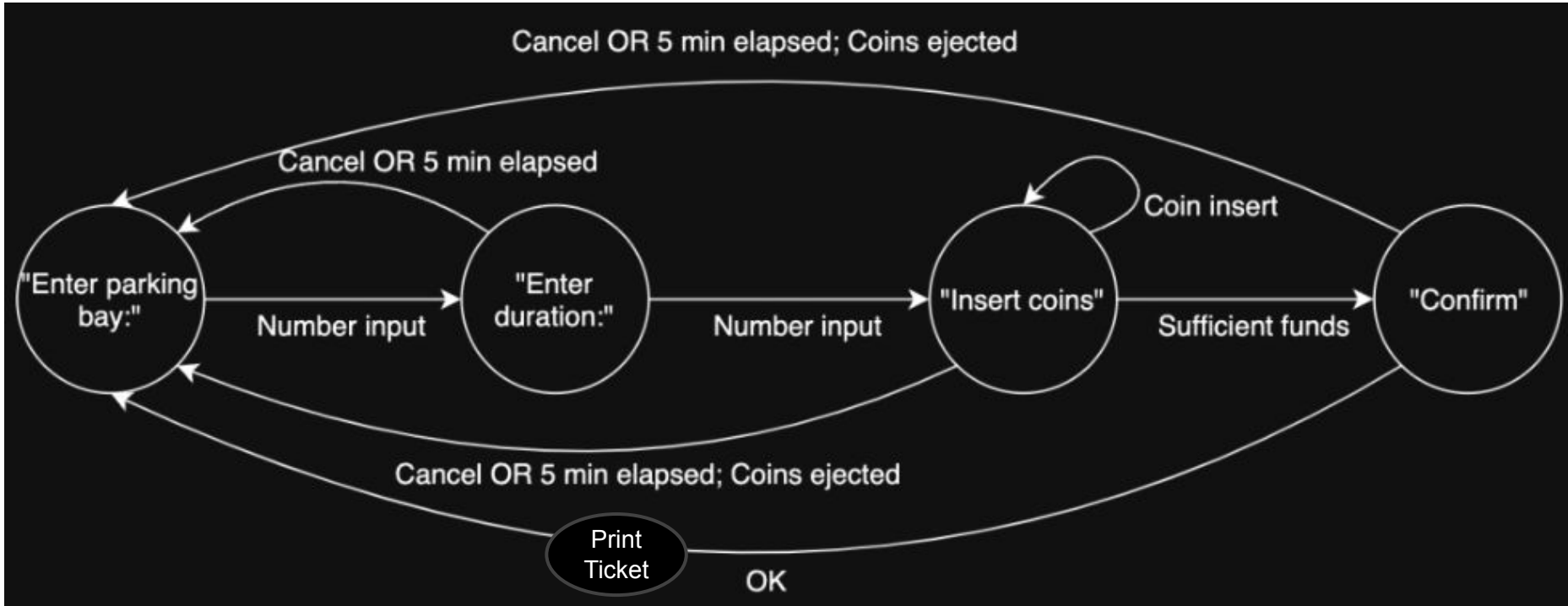


Parking Meter Example - State Machine





Parking Meter Example - State Machine





Quick Activity - Opal Card State Machine

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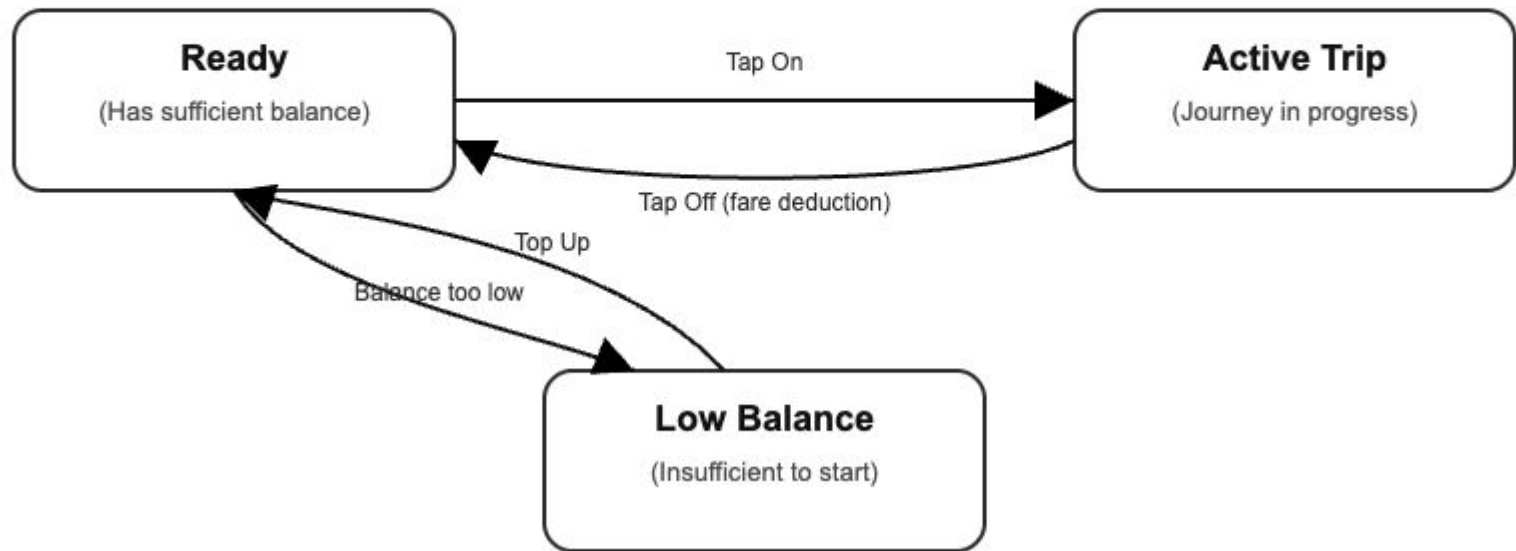
Task

In pairs or individually, **model the Opal card system** as a state machine.

Consider:

- What are the **key states** an Opal card can be in?
 - E.g. When the Card is tapped, it can read either of these states
 - i. **Ready/Idle** - *Has Sufficient balance and not yet into Journey mode*
 - ii. **Low Balance** - *System detects its balance is low*
 - iii. **Active Trip** - *Journey in Progress*
- What **actions/events** cause state changes?
 - e.g., Tap On, Tap Off, Top Up

Opal Card — Simplified State Machine



Summary

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- Models simplify complex system ideas
- Conceptual models focus on understanding
 - **Structural models** show system components (architecture)
 - **Behavioural models** show system behaviour
- State machines define valid transitions (system states) in software
- Helps design before coding begins
 - Models prevent mistakes early → saves time, effort, and cost later.

Feedback

COMP1531 Feedback Survey

